

MACHINE LEARNING PROJECT REPORT

Body Performance Analysis & Prediction

Dataset: bodyPerformance.csv | Date: March 2026 | Tasks: Classification & Regression

13,393 Records	11 Features	9 Models Trained & Compared
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1. Introduction

Physical fitness assessment has long relied on standardised battery tests that measure multiple dimensions of human performance — cardiovascular endurance, muscular strength, flexibility, and explosive power. When these measurements are collected at scale, they offer a rich ground for machine learning: the data carry natural latent structure that separates individuals by their overall fitness level, and they simultaneously encode strong quantitative signals that can be used to predict any single measurement from the rest.

This project addresses two complementary predictive tasks on the Body Performance dataset, a collection of 13,393 real physical fitness assessments recorded across age groups and genders. The first task is a four-class classification problem: given a participant's biometric and performance measurements, predict which performance tier — A (elite), B (above average), C (average), or D (below average) — they belong to. The second task is regression: predict a participant's broad jump distance in centimetres, a composite measure of lower-body power and coordination, from the remaining features.

The practical relevance of these tasks is significant. Accurate classification can assist coaches and health professionals in quickly placing athletes into appropriate training groups, while reliable regression of the broad jump score can serve as a proxy for readiness assessment when direct measurement is unavailable. The project therefore combines predictive analytics with exploratory insight, aiming not merely to achieve high accuracy but to understand what physical characteristics drive performance differences.

2. Data Preprocessing

Preprocessing was structured as a multi-stage pipeline, each stage targeting a specific source of data quality risk. Rather than applying a uniform cleaning recipe, the project made deliberate, domain-informed decisions at every step.

2.1 Initial Inspection and Type Verification

Column names were stripped of any leading or trailing whitespace to prevent silent key-lookup failures during modelling. Data types were verified against domain expectations: while age and sit-ups counts were stored as integers disguised as floats in some versions of the dataset, this was deemed acceptable since all downstream ML models handle continuous numerical inputs uniformly. The two categorical variables — gender (M/F) and class (A/B/C/D) — were identified early, as they required different encoding treatment compared to the numeric features.

2.2 Missing Value and Duplicate Checks

A systematic scan confirmed that the dataset contained zero missing values across all 12 columns, removing the need for any imputation strategy. This is noteworthy because it suggests the dataset was collected under controlled conditions where measurement was always completed. Duplicate rows were detected and removed to prevent any model from memorising repeated samples — a subtle but important source of overfitting that is often overlooked in exploratory projects.

2.3 Domain-Informed Invalid Value Filtering

Rather than relying solely on statistical thresholds, the project applied physiologically motivated validity rules to the blood pressure columns. Rows where diastolic pressure was at or below 40 mmHg, or systolic pressure at or below 70 mmHg, were removed as these values fall outside the range compatible with conscious, ambulatory participants. Critically, rows where diastolic pressure equalled or exceeded systolic pressure were also dropped — a physiological impossibility that most general outlier detection methods would miss entirely, since either value individually may appear plausible.

Additional validation checks were applied across all numeric columns using medically accepted ranges: age (12–80), height (100–250 cm), weight (30–200 kg), body fat percentage (2–60%), grip force (> 0), sit-ups counts (≥ 0), and broad jump distance (≥ 0). Of note, participants with a broad jump distance of zero were explicitly retained as valid low-performance measurements rather than treated as data errors — a nuanced decision that prevents systematic removal of the lowest-performing individuals from the dataset.

2.4 Rule-Based Outlier Removal

Rows violating the domain ranges for weight, body fat percentage, diastolic, and systolic pressure were removed outright. This two-pass approach — first correcting obvious physiological impossibilities, then removing values outside accepted clinical ranges — is more defensible than applying a purely statistical criterion such as the three-sigma rule, because it anchors decisions in what the human body can actually produce rather than in the distribution of the sample.

2.5 IQR-Based Winsorization

After rule-based cleaning, remaining outliers were handled through Winsorization using the standard IQR fencing rule ($Q1 - 1.5 \times IQR$, $Q3 + 1.5 \times IQR$). Rather than dropping these borderline records —

which would further reduce the already-filtered dataset — extreme values were clipped to the fence values. For columns whose natural minimum is zero (such as grip force and sit-ups counts), the lower fence was floored at zero to prevent the creation of negative values that would be physically meaningless. Winsorization preserves the number of training samples while dampening the leverage that extreme values would otherwise exert on distance-based and gradient-based models.

2.6 Feature Encoding and Scaling

Encoding and scaling were handled deliberately within scikit-learn Pipelines, not applied globally to the raw dataframe. This architecture choice is critical: it ensures that scaling statistics (mean, standard deviation) and encoding vocabularies are learned exclusively on training data and applied to test data, preventing any form of data leakage.

Gender, the only non-ordinal categorical variable in the feature set, was encoded using OneHotEncoder with `handle_unknown='ignore'`, producing a binary indicator column. This is appropriate given that gender has no intrinsic ordering that an ordinal scheme could exploit. The class target variable for classification was left as a string and handled natively by scikit-learn's classifiers, which perform internal label encoding. In the correlation analysis (an EDA-only step, not used for model training), gender was manually binary-encoded as a numeric indicator to enable inclusion in the heatmap.

StandardScaler was applied to all numeric features for models that are sensitive to feature magnitude — KNN, SVM, and the Neural Network. Decision Tree variants received a pass-through preprocessor that skips scaling, since trees split on individual thresholds and are invariant to monotonic transformations. Linear Regression also used the basic preprocessor because the OLS solution is scale-invariant in terms of predictions (though coefficient interpretability benefits from scaling, which was not the project's focus here).

3. Modelling Strategy

The modelling strategy was deliberately comparative rather than committed to a single approach. By training a diverse set of algorithms that make fundamentally different assumptions about the data-generating process, the project could reveal whether the underlying structure in the body performance data is linear or non-linear, local or global, smooth or discontinuous.

3.1 Classification Models

Model	Variant(s)	Key Assumption
K-Nearest Neighbours	k = 3, 5, 7	Performance class is determined by local neighbourhood similarity
Decision Tree	max_depth = 3, 5, None	Class boundaries are axis-aligned and hierarchical
Support Vector Machine	Linear kernel, RBF kernel	Class boundaries are hyperplanes (linear) or curved (RBF)
Neural Network (MLP)	Layers: 64 → 32, ReLU	Complex non-linear mappings via learned representations

K-Nearest Neighbours (k = 3, 5, 7)

KNN was chosen as a non-parametric baseline. It makes no assumptions about the functional form of the class boundaries and classifies each sample purely by the majority vote of its nearest neighbours in feature space. Three values of k were tested to observe the bias-variance trade-off: smaller k is more flexible and sensitive to local noise, while larger k smooths decision boundaries at the cost of fine-grained discrimination. Because KNN distances are dominated by features with large scales, it requires standardised inputs.

Decision Tree (depth = 3, 5, None)

Decision Trees were included because they produce explicitly interpretable rule sets — a practical advantage if the model needs to be explained to non-technical stakeholders such as coaches. The three depth configurations span a wide complexity range: depth 3 produces an extremely simple model that can be visualised on a single page; depth 5 allows moderate complexity; and unconstrained depth captures all training structure, deliberately inviting overfitting to serve as an upper bound on training-set fit.

Support Vector Machine (Linear and RBF)

Two SVM kernels were compared to explicitly test whether the class boundaries are linearly separable in the original feature space. The linear kernel is the simpler hypothesis; the RBF kernel (with gamma='scale') allows curved, non-linear separation. This pairing provides a direct, controlled comparison of linear versus non-linear decision boundaries under an otherwise identical modelling framework.

Neural Network (MLP)

The MLP with a 64→32 hidden layer architecture was included as the most expressive model in the classification set. Its ability to learn hierarchical feature interactions makes it a natural candidate for

a dataset where physical performance likely involves complex interactions — for example, the combined effect of age, grip force, and flexibility on overall class assignment. With ReLU activations and 500 maximum iterations, it trades interpretability for representational power.

3.2 Regression Models

Model	Variant(s)	Key Assumption
Linear Regression	OLS	Broad jump distance is a linear combination of features
K-Nearest Neighbours Regressor	k = 3, 5	Broad jump is predicted by local averaging of neighbours
Decision Tree Regressor	max_depth = 3, 5	Jump distance follows piecewise-constant regions in feature space
Support Vector Regressor	Linear, RBF	Regression within an epsilon-insensitive tube
Neural Network Regressor	Layers: 64 → 32, ReLU	Non-linear function approximation

Linear Regression was retained as the interpretable baseline. If the data were purely linear, it would perform competitively while offering full coefficient transparency. The remaining models mirror the classification choices, enabling a consistent comparison across both tasks and revealing whether the broad jump prediction problem shares the same structural complexity as the classification problem.

4. Pipeline & Workflow

Every model in this project — across both classification and regression — was constructed as a scikit-learn Pipeline object. Each Pipeline encapsulates a ColumnTransformer for preprocessing followed by the estimator, treating the two stages as a single, atomic unit during training, validation, and prediction.

The consequences of this design go beyond code cleanliness. When `cross_val_score` is called on a Pipeline, scikit-learn splits the data first and then calls `fit` on the pipeline using only the training fold. The ColumnTransformer — including the StandardScaler's mean/std computation and the OneHotEncoder's vocabulary — is fitted exclusively on the training fold data. Test fold data are then transformed using these training-fold statistics. This guarantees that no information from the test set contaminates the preprocessing, which is the most common but least visible form of data leakage in ML pipelines.

Two distinct preprocessors were constructed: a scaled version (StandardScaler + OneHotEncoder) for distance- and gradient-sensitive models (KNN, SVM, MLP), and a basic version (passthrough + OneHotEncoder) for tree-based models and linear regression. This differentiation is intentional: applying standardisation to tree models is harmless but unnecessary, while failing to apply it to KNN or SVM would produce silently degraded results as features measured in kilograms would dominate features measured in centimetres.

The train/test split experiments used stratified splitting for classification (`stratify=y_cls`), ensuring that the class distribution A:B:C:D is preserved in both training and test sets regardless of the split ratio. Three split ratios — 80:20, 70:30, and 50:50 — were evaluated to assess sensitivity to training set size. The regression splits used a fixed `random_state=42` without stratification, which is standard for continuous targets. The K-Fold cross-validation phase used `StratifiedKFold` (5 splits) for classification and standard `KFold` (5 splits) for regression, with `shuffle=True` and `random_state=42` to ensure reproducibility.

5. Model Evaluation

5.1 Classification Metrics

The primary ranking metric for classification was macro-averaged F1 score. This choice reflects a deliberate judgment about what matters in this problem. The dataset is approximately balanced across the four performance classes, so accuracy alone would be a reasonable metric — but macro F1 provides an additional guarantee: it weights each class equally regardless of size, ensuring that the model is not artificially rewarded for strong performance on the most frequent class at the expense of minorities.

Alongside F1, macro-averaged precision and recall were reported. Precision captures the cost of false positives — incorrectly labelling a person as being in a higher performance tier than they actually occupy. Recall captures the cost of false negatives — missing individuals who belong to a given performance class. In a fitness assessment context, both errors carry real consequences: overestimation leads to inappropriate training load, while underestimation may cause undertraining and missed development. Reporting both gives a complete picture of the error profile.

The confusion matrix was generated for the best-performing model trained on the full dataset, providing a granular view of where class confusions occur — specifically, whether the model tends to confuse adjacent classes (e.g., B vs C) or distant ones (e.g., A vs D), which would carry different practical implications.

5.2 Regression Metrics

Three metrics were computed for each regression model: Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and the coefficient of determination (R^2). MSE is the optimisation criterion used internally by OLS and most gradient-based regressors, making it a natural reference point. RMSE is the square root of MSE and has the advantage of being expressed in the same units as the target variable (centimetres of broad jump distance), making errors directly interpretable — an RMSE of 10 cm means the model's predictions are off by roughly 10 cm on average in a root-mean-squared sense.

R^2 serves as the primary ranking criterion, as it expresses the fraction of variance in broad jump distance explained by the model, normalised to a 0–1 scale (with negative values possible for models worse than a naive mean predictor). An R^2 of 0.80, for instance, means the model accounts for 80% of the variability in broad jump distance, leaving 20% unexplained by the included features. The primary ranking metric for the cross-validation phase was also R^2 , enabling consistent comparison across all regression models.

6. Results & Insights

6.1 Classification Results

Across the three train/test split experiments and the 5-fold cross-validation phase, the SVM with RBF kernel and the Neural Network (MLP) consistently emerged as the top-performing classification models. The KNN variants (particularly $k=5$) also produced competitive results, typically outperforming the Decision Tree configurations.

The unconstrained Decision Tree (`max_depth=None`) exhibited a characteristic pattern: near-perfect or perfect accuracy on the training set, followed by a meaningful performance drop in cross-validation. This is the textbook signature of overfitting — the model memorised the training data rather than learning generalisable patterns. The depth-3 Decision Tree, by contrast, underfit: its deliberately shallow structure could not capture the complexity needed to discriminate four performance tiers.

The SVM with a linear kernel performed noticeably below its RBF counterpart, which is a meaningful finding. It suggests that the boundaries separating performance classes A through D are not linearly separable in the original 11-dimensional feature space — there exist interactions between features (for example, the combination of age, grip force, and body fat percentage) that define class membership in ways that a hyperplane cannot capture. The RBF kernel, by implicitly projecting data into a higher-dimensional space, can accommodate these non-linear boundaries and therefore achieves substantially better separation.

The Neural Network's competitive performance corroborates this finding: both the RBF SVM and the MLP succeed precisely because they can represent non-linear decision surfaces. The fact that they perform similarly suggests that the complexity required is moderate — captured adequately by both a kernel method and a two-layer neural network — rather than demanding the very deep architectures needed for genuinely high-dimensional problems.

6.2 Regression Results

For the regression task — predicting broad jump distance — the Neural Network Regressor and SVR with RBF kernel again led the rankings in cross-validation R^2 scores, consistent with their classification counterparts. The Linear Regression baseline achieved a respectable R^2 that was meaningfully below the top non-linear models, confirming that broad jump distance is not a simple additive function of the input features.

This finding is interpretable in light of the EDA. The correlation heatmap revealed that grip force and broad jump distance are strongly positively correlated (~ 0.73), while age and broad jump show a moderate negative correlation (~ -0.50). However, correlation measures only pairwise linear relationships. The regression results suggest that the true predictive signal involves interaction effects — for example, the joint influence of age and grip force may be more predictive than either alone — which linear regression handles poorly but tree-based and kernel-based models can accommodate.

The Decision Tree Regressor with depth 5 produced a moderate R^2 , falling between the linear baseline and the top non-linear models. This is consistent with its ability to capture some non-linearity through recursive partitioning, but its piecewise-constant prediction surface introduces an approximation error that continuous models do not.

6.3 Cross-Validation Consistency

The use of 5-fold cross-validation as the final ranking criterion was critical because the three train/test split experiments revealed that model rankings are not entirely stable across different split ratios — particularly at the 50:50 split where reduced training data amplifies variance. The cross-validation results, by averaging across five independent splits, provide a more reliable estimate of generalisation performance. The stability of the top models (SVM-RBF and MLP) across all evaluation configurations increases confidence that their performance reflects a genuine signal rather than a favourable random split.

7. Critical Analysis

7.1 Limitations

- The best-model evaluation in Part 3 was performed by re-training on the entire dataset and predicting on the same data. This produces training-set metrics that are optimistically biased — not genuine out-of-sample estimates. A proper hold-out test set should have been set aside at the very beginning of the project and used only once, at the final evaluation stage.
- Hyperparameters for all models were fixed at plausible defaults (e.g., $C=1.0$ for SVM, $k=5$ for KNN, two hidden layers of 64 and 32 neurons for MLP) rather than tuned through grid or random search. This means the reported performances represent each model's performance at an arbitrary configuration, not its optimal configuration. The relative ranking of models may differ after proper tuning.
- The neural network was run for a maximum of 500 iterations without monitoring validation loss curves. There is no confirmation that the MLP converged, and without an explicit validation set during training, early stopping could not be applied to prevent overfitting during gradient descent.
- The regression target — broad jump distance — was also used as a feature in the classification pipeline (since it is one of the 11 measurements). While this does not constitute leakage in the strict sense (both tasks use different target variables), it means the classification model partially relies on the very variable that the regression model aims to predict. If this system were deployed for real prediction, the two tasks would need to be decoupled carefully.
- The dataset, while reasonably large at ~13,000 records, reflects a specific population captured under particular testing conditions. Generalisation to populations with different demographics, fitness cultures, or testing protocols is not guaranteed by the current evaluation.

7.2 Strengths

- The preprocessing pipeline architecture using scikit-learn's Pipeline and ColumnTransformer is methodologically sound and prevents data leakage in all cross-validation experiments.
- The use of domain knowledge to filter physiologically impossible values (inverted blood pressure, zero grip force) demonstrates statistical literacy combined with subject-matter reasoning.
- Winsorization was preferred over outlier deletion, preserving dataset size and avoiding the selection bias introduced by wholesale removal of extreme performers.
- Multiple split ratios were tested, providing insight into how sensitive model rankings are to training set size — a check that single-split evaluations cannot provide.

8. Conclusion

This project successfully developed and evaluated a complete machine learning workflow for two distinct predictive tasks on a body performance dataset. The preprocessing phase demonstrated that thoughtful, domain-informed cleaning — including physiologically motivated filtering, rule-based outlier removal, and IQR Winsorization — produces a high-quality dataset without unnecessarily discarding informative samples.

Across both classification and regression tasks, the evidence consistently points to non-linear models as the most appropriate choice for this data. The SVM with RBF kernel and the Neural Network (MLP/MLPRegressor) outperformed their linear and simpler counterparts across multiple evaluation configurations. This convergence of evidence from two independent kernels, two evaluation methodologies (hold-out splits and cross-validation), and two tasks (classification and regression) is a strong signal that the physical performance data contains genuine non-linear structure — specifically, interactions between age, strength, and flexibility that linear models cannot represent.

For the classification task, the SVM-RBF and Neural Network are jointly recommended as the final models, with the SVM offering a slight edge in interpretability (the support vectors carry structural meaning) and the MLP offering greater flexibility for future architectural expansion. For regression, the same models lead, with the Neural Network Regressor being the primary recommendation given its slightly greater capacity for learning complex functional relationships.

The Decision Tree variants, while intuitively appealing for their human-readable output, demonstrated a narrow window between underfitting (depth 3) and overfitting (unconstrained), and did not match the top performers in cross-validation. They may still serve a valuable supplementary role — providing explainable rule sets for stakeholder communication — but should not be used as the primary predictive model.

9. Future Improvements

9.1 Hyperparameter Optimisation

The most immediate improvement would be to replace fixed hyperparameter configurations with a systematic search. For the SVM, the regularisation parameter C and kernel coefficient γ are both highly influential and were left at defaults. For the MLP, network depth, layer width, learning rate, and regularisation strength are all tunable. A `RandomizedSearchCV` with 5-fold inner cross-validation (nested CV) would provide reliable, unbiased estimates of the optimal configurations without overfitting to the validation set.

9.2 Ensemble Methods

Random Forest and Gradient Boosting (XGBoost, LightGBM) were not included in this project but represent the most natural next step. Random Forests reduce the Decision Tree's overfitting problem by averaging many independently grown trees, typically producing strong out-of-box performance on tabular data. Gradient Boosting models are frequently the top performers on structured datasets in competitive settings and would serve as a meaningful benchmark against the SVM and MLP.

9.3 Feature Engineering

The project used raw measurements directly. Several derived features could carry additional predictive signal — for example, BMI ($\text{weight}/\text{height}^2$) as a composite of two existing features, or a strength-to-weight ratio ($\text{grip force} / \text{weight_kg}$). Age-stratified normalisation of performance metrics (expressing each measurement relative to age-group norms) could also remove the confounding effect of age and reveal residual performance differences that are more informative for class prediction.

9.4 Proper Hold-Out Test Set

A clean test set — set aside before any preprocessing or modelling decisions — should be established and used only for the final reported metrics. This eliminates the risk that iterative preprocessing choices (e.g., choosing Winsorization over deletion after observing model performance) have inadvertently adapted the pipeline to the "test" data.

9.5 Explainability Layer

For a dataset with direct human health implications, model explainability is not optional. SHAP (SHapley Additive exPlanations) values can be computed efficiently for both the SVM and tree-based models to quantify each feature's marginal contribution to individual predictions. This would allow practitioners to understand, for instance, whether a particular athlete was classified as Class D primarily due to age, low grip force, or high body fat — information that directly guides targeted intervention.

9.6 Class-Specific Threshold Optimisation

The current classification models use default decision thresholds (argmax of predicted class probabilities). For an SVM or calibrated MLP, class-specific thresholds could be tuned using

Precision-Recall curves to trade off precision and recall according to the relative cost of each type of error — for instance, accepting lower precision for Class A (fewer false positives) in exchange for higher recall (catching more true elite performers).